RCM & Billing Decision Guide

Use this guide to decide which parts of the revenue cycle your application owns and which it delegates to billing partners. Work through coverage, charge capture, claims, remittance, and patient responsibility to define the data model, integrations, and operational handoffs for your reimbursement model.

Companion to the Billing docs.

Section 1: Use Case & Participants

1.1 What is your reimbursement model?

- Insurance / claims-based – you bill payers and reconcile their responses
- Patient self-pay / cash – patients pay you directly; no payer claim
- Value-based / capitation – per-member-per-month or risk-based
- Some combination

Why: Claims-based reads every lane; self-pay skips the Charge → Claim → Payer lane; value-based leans on Foundations and the Value-Based lane (3.11).

1.2 How much of the revenue cycle runs in Medplum vs. delegated?

- **Clearinghouse-direct** – Medplum runs the cycle; the clearinghouse is transport. You own scrubbing, denials, and remittance posting. The most complete documented path.

- **RCM-automation partner** – you send encounter and charge data and delegate the back end; scrubbing, payer routing, denials, resubmission, patient collections, and reporting run on the partner's platform. The integration surface is claim submission; the partner's billing tasks sync back into Medplum as `Task`s.
- **Documents only (no electronic claims)** – you don't run claims through a clearinghouse or RCM partner. Generate a CMS-1500 PDF for the occasional paper/portal payer, or hand patients a superbill to self-submit for out-of-network reimbursement. Common for cash-pay and OON practices (see 1.1); also a per-payer fallback you can layer onto any engine above.
- **External Practice Management (PM) / billing system** – Medplum is the clinical source of truth; charges or claims hand off to an existing PM/billing system that owns the rest.

In one line: a clearinghouse moves your claims; an RCM platform runs your revenue cycle. Either way Medplum owns the front half (coverage, charge capture, claim assembly); the paths diverge on the back half.

Revenue-cycle stage	Clearinghouse	RCM partner	Documents only	External PM
Coverage & charge capture (3.1–3.5)	You build	You build	You build	You build
Eligibility (3.3)	You build	Partner platform	Manual	External
Claim assembly (3.6)	You build	You build	You build	External
Scrubbing / validation	You build	Partner owns	Light	External
Submission (3.6)	You build (837P)	Partner submit op	PDF / patient files	External

Revenue-cycle stage	Clearinghouse	RCM partner	Documents only	External PM
Responses & ERA posting (3.7)	You build (webhook + poller)	Partner owns; Tasks sync back	Manual	External
Denials & resubmission (3.8)	You build	Partner owns; Tasks sync back	Manual	External
AR & reporting (3.10)	You build	Partner owns	Manual	External

How to choose:

	Clearinghouse	RCM partner
You get	Full control; billing data stays as FHIR in Medplum	A payer-rules engine and in-house billers; far less to build
Trade-off	You build and maintain the entire back office	Recurring fees; less control; AR and reporting live on the partner's platform
Choose if	You have RCM expertise and want a fully owned cycle	You want to outsource the cycle

Paths can mix. The choice sets which Section 3 lanes you build (3.5–3.10) vs. read only for the boundary.

1.3 Who does the billing work, and is the patient part of it?

- Charge capture: clinicians at point of care, or back-office entry?
- Coding and claim work: dedicated coders/billers, or automated with exception review?
- Accounts receivable (AR) follow-up: who works denials, aging, and patient balances?
- Patient: do patients see estimates, statements, or pay online?

Why: drives charge-capture design (3.5), who works denials and AR (3.8, 3.10), and whether patient-facing billing is in scope (3.9).

1.4 Where does coding happen, and how automated is it?

- Auto-derived from clinical data (visit type, orders, procedures)?
- Coder-reviewed before the claim is built?
- Fully manual?

Why: drives the coding and review model in charge capture (3.5) and where scrubbing exceptions surface (3.6).

1.5 New build or replacing/extending, and what else must billing touch?

- If replacing: what billing/PM system, and what gaps drove the change?
- If new: is billing core to the product (you're building an RCM/billing product) or additive (a clinical app that also needs to bill)?
- What other systems must billing read from or write to (clinical/EHR source, PM, clearinghouse, accounting/GL, payment processor, payer portals)?

Why: surfaces integration boundaries and the source-of-truth split behind 1.2, 3.6, and 3.7. Core-to-product builds tend to own the cycle (a clearinghouse); additive builds tend to delegate (an RCM partner or external PM).

Section 2: Feature Scoping

Go through each row together. For each cell under Yes / No / Nice-to-have / Not sure, mark the customer's answer. The **§** *column points to the deep-dive subsection that covers each feature.*

#	Feature	§	Yes	No	Nice-to-have	Not sure
1	Coverage & payer modeling	3.1				

#	Feature	§	Yes	No	Nice-to-have	Not sure
2	Patient financial account / running balance	3.2				
3	Real-time eligibility & benefits (270/271)	3.3				
4	Prior authorization (pre-service)	3.4				
5	Patient cost / Good-Faith Estimates	3.4				
6	Charge capture (encounter-linked) & pricing / fee schedules	3.5				
7	Coding (CPT/ICD-10, modifiers) with review queue	3.5, 3.6				
8	Assemble, validate & submit claims	3.6				
9	CMS-1500 PDFs / superbills (manual / patient submission)	3.6				
10	Ingest payer responses & post remittance (277CA / 835)	3.7				
11	Denials, corrected claims & appeals	3.8				
12	Patient statements, invoices & online payment	3.9				

#	Feature	§	Yes	No	Nice-to-have	Not sure
13	Claim / AR status, aging & reporting	3.10				
14	Value-based contracts (panel, quality measures, capitation)	3.11				

Section 3: Feature Deep Dives

Cover each feature flagged Yes or Nice-to-have in Section 2. The goal is to land on a clear recommended approach by the end of each section.

Section 3 is grouped into four lanes:

Lane	Subsections	When to read
Foundations	3.1 – 3.2	Always
Pre-service / Patient Access	3.3 – 3.4	When you verify coverage or authorize before service
Charge → Claim → Payer	3.5 – 3.8	When you bill insurance
Patient & Operations	3.9 – 3.10	Always
Value-Based	3.11	When you have capitated or risk-based contracts

Foundations

3.1 Coverage & Insurance Model

How you represent a patient's insurance (the coverage stack, the subscriber, payers, cost-sharing) is the foundation every claim and eligibility check builds on. See Representing Patient Insurance Coverage; coverage capture at intake is in the Intake guide (3.5).

Questions:

- Do patients commonly have more than one coverage, and must you bill them in order?
- Is the subscriber sometimes someone other than the patient?
- Is your payer list curated in Medplum, synced, or created on demand?
- What cost-sharing (copay, coinsurance, deductible) do you need for estimates and patient billing?

Situation	Approach
Subscriber is the patient	One `Coverage`; `subscriber` = `Patient`; `relationship` = `self`; `payor` → `Organization`.
Primary + secondary	One `Coverage` per plan; `Coverage.order` sets billing sequence. Secondary submits after the primary remittance posts (3.7).
Subscriber ≠ patient	`Coverage` + `RelatedPerson` for the subscriber. The relationship inverts: `Coverage.relationship` is the patient's relationship to the subscriber, the `RelatedPerson`'s is the reverse (coverage `child` → related person `parent`). Getting this backwards is a common billing bug.
Member ID & plan details	`Coverage.subscriberId` for member ID; `Coverage.class` for group, plan name, RxBIN/RxPCN.
Cost-sharing	`Coverage.costToBeneficiary`; feeds estimates (3.4) and patient responsibility (3.9).
Payer directory	Medplum ships a built-in **Payer Directory** of `Organizations`, or curate your own. The payer identifier system differs by backend; see 3.6.

Situation	Approach
Card images	Store as `DocumentReference`/`Binary` linked to the `Coverage`.

3.2 Account & Financial Ledger

Decide whether you track a patient financial account that rolls charges, claims, and payments into a balance (the backbone for AR in 3.10 and patient billing in 3.9), or let each charge stand alone.

Note: the rows below are one FHIR approach to a financial ledger; validate against your reporting and reconciliation requirements.

Questions:

- Do you need a single balance rolling up charges, payer payments, adjustments, and patient payments?
- Is the account scoped per patient, per visit, or per episode?
- Do you track payer and patient responsibility separately on the same account?
- Must the account reconcile against an external accounting/GL system?

Situation	Approach
Running balance	`Account` as the ledger anchor; `ChargeItem`, `Claim`, `PaymentReconciliation`, and `Invoice` reference it.
Account scope	`Account.subject` → `Patient`, or tie to `Encounter` for visit-level; apply one scope consistently.
Payer vs. patient	Payer side via claim/remittance (3.7), patient side via `Invoice` (3.9), both rolling to the same `Account`.
No ledger	Skip `Account` for pure pass-through submission where another system owns the balance, but you lose in-Medplum AR (3.10).

Pre-service / Patient Access

3.3 Eligibility & Benefits

Decide whether you verify coverage before service, what you ask the payer, and when the check runs. See Insurance Eligibility Checks.

Note: Eligibility is provided by the clearinghouse integration. An RCM partner may offer its own eligibility on its platform, outside that integration.

Questions:

- What must you confirm: active coverage, covered service, in-network status, or benefit/cost detail?
- When does the check run: scheduling, check-in, or batch pre-visit?
- Do you re-check on a cadence for ongoing patients?
- Who works failed or ambiguous checks?

Situation	Approach
Verify coverage / benefits	`CoverageEligibilityRequest` (`purpose: benefits`); invoke the clearinghouse eligibility operation to run the X12 270/271 and return a `CoverageEligibilityResponse`.
Scope limit	The current integration returns general benefits only, not arbitrary service-specific benefit queries.
In-network check	Read network participation from the response (in-network indicator Y/N/U/W); run at provider selection or as a pre-schedule gate.
Cost-sharing	Copay, coinsurance, remaining deductible, and out-of-pocket land in `CoverageEligibilityResponse.insurance.item`; the input to patient estimates (3.4).

Situation	Approach
When it runs	A `Bot` issues the request on the triggering event; persist the response for the visit and billing.
Exception	On failure or ambiguity, create a `Task` for registration rather than proceeding silently.

3.4 Prior Authorization & Estimates

Decide whether services need payer authorization before they happen, how you track that to a decision, and whether you produce patient cost estimates up front.

Note: the eligibility- and Claim-based rows below are one FHIR approach to prior auth; validate against your payer requirements. The standards-based CMS-0057-F path (CDS Hooks / CRD) is in alpha. Prior auth in a referral context is in the Referrals guide (3.7).

Questions:

- Do any services require prior authorization before scheduling or delivery?
- How do you check whether auth is required, and track a submitted auth to approval/denial?
- Must scheduling be gated until authorization clears?
- Do you owe patients a cost estimate before service (e.g. a Good-Faith Estimate)?

Situation	Approach
Is auth required?	`CoverageEligibilityRequest` with `purpose: auth-requirements`; the response indicates whether authorization is required.

Situation	Approach
Submit a prior auth	`Claim` with `use: preauthorization` (distinct from a claim for completed services); `ClaimResponse` carries the decision. A `Bot` converts FHIR ↔️ X12 for transmission.
Standards-based (CMS-0057-F)	Electronic prior auth via CDS Hooks / Coverage Requirements Discovery (CRD); alpha. The emerging payer-interop path as the rule phases in.
Gate scheduling	Hold in an auth-pending `Task.businessStatus` until the `ClaimResponse` approves, then release.

Good-Faith Estimates (No Surprises Act). One FHIR approach: build the estimate from pricing rules (`ChargeItemDefinition`/`$apply`, 3.5) and the patient's cost-sharing (`Coverage.costToBeneficiary`, 3.1), delivered as a `DocumentReference`; validate against your compliance requirements. Mainly for self-pay and out-of-network patients.

Charge → Claim → Payer

Insurance-claims-specific. Self-pay builds skip to Patient & Operations; pricing (3.5) still applies to self-pay rates and estimates.

3.5 Charge Capture, Coding & Pricing

Decide what gets billed, where the codes come from, and how each line is priced. Coding is the largest source of denials, so treat it as first-class, not a byproduct of charge entry.

Questions:

- What generates a charge: a completed encounter, an order/procedure, a fulfilled service?

- Where do codes come from (CPT/HCPCS, ICD-10, modifiers): auto-derived, coder-reviewed, or manual (1.4)?
- Is pricing a single fee schedule, or does it vary by payer contract?
- Who reviews coding before a claim is built, and how do exceptions surface?

Situation	Approach
What gets billed	One `ChargeItem` per billable line; `ChargeItem.code` carries CPT/HCPCS. Link `context` → `Encounter` and reference the `Account` (3.2) at creation, or reconciling into claims and AR later gets painful.
Pricing	`ChargeItemDefinition` holds the rules; `$apply` calculates the price.
Payer-specific rates	Multiple `ChargeItemDefinitions` keyed to payer/contract; apply the one matching the coverage.
Coding source	Auto-derive via a `Bot`, or set by a coder; route low-confidence items to a review `Task`.
Diagnoses	Capture ICD-10 diagnoses to attach to the claim (`Claim.diagnosis`, with item-level pointers) in 3.6.

3.6 Claim Assembly & Submission

Decide how you build a valid claim from charges and coverage, and how it leaves your system. The 1.2 engine decision plays out here.

⚠️ **One-way door:** Choosing a submission backend is hard to reverse. Each maps the `Claim` to a different format (837P, the partner's API, a CMS-1500 PDF) with a different response pipeline (3.7), and payer identifier systems differ. Keep one `Claim` as the source of truth so the backend stays swappable.

Questions:

- Are you billing as an organization or an individual provider?

- Which submission path(s), and does it vary by payer (electronic for most, paper/portal for some)?
- What validation must pass before submission, and where do failures surface?
- Do you submit secondary claims after the primary pays (coordination of benefits)?

Situation	Approach
Build the claim	`Claim` referencing the `Coverage` (`Claim.insurance`), charges (`Claim.item`), and ICD-10 diagnoses (`Claim.diagnosis`) with item-level pointers.
Billing provider	Set on `Claim.provider`: the billing `Organization` (or a `Practitioner` for individual billing). The rendering provider goes on `Claim.careTeam` with role `primary`.
Electronic (you own the cycle)	The clearinghouse submit operation sends an 837P and writes a correlation id back to the `Claim`.
RCM partner	The partner's submit operation returns a `ClaimResponse`; the partner owns scrubbing, denials, resubmission, and remittance on its platform, and billing tasks sync back into Medplum as `Tasks` (3.8).
Manual / paper / portal	`Claim/$export` generates a CMS-1500 PDF (`Binary`; experimental); log the outbound document.
Patient-submitted (out-of-network)	Generate a superbill for the patient to file.

Situation	Approach
Validation	Validate before submit; surface a `Task` and don't transmit invalid claims. (In the current integration, a submission failure marks the `Claim` `status = error`.)
Multiple channels	One `Claim` is the source of truth; run separate send paths per payer without diverging content.

3.7 Payer Responses & Remittance Posting

Decide how you ingest what the payer sends back (the *277CA* acknowledgment, accepted or rejected pre-adjudication, and the *835 ERA* remittance, what was paid, adjusted, or denied line by line) and how payments post. These share one pipeline, so design them together.

Note: applies to the clearinghouse path, and the row specifics reflect the current integration. With an RCM partner, the partner owns response handling and posting; what flows back is its billing `Tasks`.

Questions:

- How do responses arrive: webhook push, polling, or partner API?
- How do you match a response to the originating claim?
- How do payer payments and adjustments post to the account?
- What share posts automatically vs. needs review?

Situation	Approach
Ingestion	A webhook bot (primary) plus an optional poller (catch-up). Make it idempotent on the payer's inbound transaction id (not the webhook delivery id); webhooks redeliver, and without dedup you double-post payments.

Situation	Approach
Acknowledgment (277CA)	Persist as a `ClaimResponse`; surface front-end rejections to denials (3.8).
Remittance (835 ERA)	Create a `PaymentReconciliation` and `ClaimResponse`; the ERA PDF is stored as a `DocumentReference`. Post payments/adjustments against the `Account` (3.2).
Matching	Match on the patient-control-number (PCN) assigned at submission from the `Claim` id (835), or the correlation id (277).
Claim status	Status lives on `ClaimResponse`; query `ClaimResponse?request=Claim/{id}` (3.10).
Review	Auto-post clean remittances; route mismatches to a `Task`.

3.8 Denials, Rejections & Resubmission

Decide how you work claims the payer rejects or denies: correcting and resubmitting, or appealing.

Note: The rows below are one FHIR approach to denial handling; validate against your workflow. On the clearinghouse path you build this in Medplum; with an RCM partner, denials are worked on the partner platform and surface back as synced Tasks.

Questions:

- Do you work front-end rejections (277CA) differently from adjudicated denials (835)?
- When you fix and resend, is it a corrected claim or a fresh submission?
- Do you track appeals (deadlines, supporting documentation, outcome)?
- How do staff find claims that need work?

Situation	Approach
Surface denials	Denial detail lives on the `ClaimResponse` (3.7); create a `Task` work queue for the biller.
Corrected claim	Resubmit a corrected `Claim` linked to the original (`Claim.related`), via the same backend (3.6).
Appeals	Track as a `Task` with deadlines; attach supporting docs as `DocumentReference`.
Find work	Search over open denial `Tasks` (3.10).
Partner-owned	The RCM partner works denials/resubmission on its platform; surfaced tasks sync into Medplum as `Tasks`.

Patient & Operations

3.9 Patient Responsibility & Self-Pay

Decide how you bill patients for their share (copays, coinsurance, deductibles, full self-pay) and how they pay.

Questions:

- What balances do you bill: post-adjudication responsibility, self-pay, or both?
- Do you issue statements, and on what cadence?
- Do you take online payment / card on file, through which processor?
- For out-of-network patients, do you provide a superbill instead of billing the payer?

Situation	Approach
Patient balance	`Invoice` for patient-owed amounts rolling to the `Account` (3.2); responsibility from remittance (3.7) or self-pay pricing (3.5).

Situation	Approach
Online payment	Integrate a payment processor; a sample integration shows keeping invoices and payments in sync.
Self-pay / cash	Price via `ChargeItemDefinition/$apply` (3.5); invoice the patient directly, no payer claim.
Out-of-network	Provide a superbill (3.6) for the patient to submit for reimbursement.

3.10 Accounts Receivable, Aging & Lifecycle Status

Decide how you track where every claim and balance sits, and how staff find work that's stuck or overdue.

Note: The rows below are one FHIR approach to AR tracking; validate against your reporting needs. Most relevant on the clearinghouse path; an RCM partner owns AR and reporting.

Questions:

- What lifecycle states matter (e.g. draft → submitted → accepted → paid → denied → closed)?
- How do you standardize states so queues, reports, and integrations agree?
- Do you track aging (days since submission, days pending) and AR buckets?
- How do staff find overdue or stuck claims and balances?

Situation	Approach
Lifecycle states	Track progress with `Task.businessStatus`; move `Task.status` on clear transitions. Keep the state set small and single-sourced.
External signals	`Bots`/subscriptions advance status when responses arrive (3.7): accepted, paid, denied.

Situation	Approach
Aging & AR	Search over `Claim.created`, `Task.lastModified`, and `Account` balances; bucket by age.
Overdue work	Scheduled `Bots` or subscriptions flag items past an SLA into work queues.

Value-Based

3.11 Value-Based & Capitation

Value-based care isn't a claims pipeline; it splits into measurement (quality, gaps in care) and money (capitation, risk, settlement). Read only for capitated or risk-based contracts.

Note: Quality measurement uses documented operations; the financial rows are one FHIR approach to validate against your contracts. Quality/HEDIS reporting overlaps clinical analytics; see Analytics.

Questions:

- Which contracts are capitated or risk-based, and which members are attributed to you?
- Which quality measures (eCQM, HEDIS, stars) are you accountable for?
- Do you still submit claims or encounter data when capitated (for risk adjustment and measurement)?
- How are capitation payments and any shared-savings settlement reconciled?

Situation	Approach
Attribution / panel	`Group` as the member roster; ingest the payer roster via a `Bot`; `CareTeam` for accountable teams.
Quality measurement & gaps in care	`Measure` + `MeasureReport` via `Measure/$evaluate-measure`; eCQM and HEDIS via Analytics.

Situation	Approach
Risk adjustment (HCC / RAF)	Capture `Condition` coding from encounters and submit as encounter data; ties to coding (3.5).
Encounter data	Even when capitated, reuse claim assembly (3.6) to send 837/encounter data for measurement and risk.
Payments (PMPM, settlement)	Model received capitation and shared-savings settlement as `PaymentReconciliation` against the `Account` (3.2).

Related Integrations

The billing and revenue-cycle partners Medplum integrates with first-party:

- **Stedi** — X12 eligibility checks (270/271) and professional claim submission (837P) with claim responses.
- **Candid Health** — revenue cycle automation and professional claim submission from FHIR `Claim` resources.